13 –Revenue Maximization and MNL Choice Model

IEOR 145/245

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Maximize Expected Revenue

1. Maximize probability that something is chosen

$$\max_{S \subseteq [n]} \phi(S)$$

2. Maximize expected revenue from choice

$$\max_{S \subseteq [n]} \sum_{i \in S} r_i \, \phi(i, S)$$

Objectives are identical when prices are identical

Revenue is Not Monotone

$$\max_{S \subseteq [n]} R(S)$$

where $R(S) = \sum_{i \in S} r_i \, \phi(i, S)$

- 2 products, RCM with $\lambda_{1\,2\,0} = \lambda_{2\,0\,1} = 0.5$
- $r_1 = 1, r_2 = 2$
- $R(\{2\}) = 2 > 1.5 = R(\{1,2\})$
- This effect is called “cannibalization”

Revenue is Not Submodular

$$\max_{S \subseteq [n]} R(S)$$

where $R(S) = \sum_{i \in S} r_i \, \phi(i, S)$

- 3 products, RCM with $\lambda_{3\,1\,2\,0} = \lambda_{2\,3\,0\,1} = 0.5$
- $r_1 = 1, r_2 = 2, r_3 = 2$

- $R(\{3,1,2\}) - R(\{1,2\}) = 2 - 1.5 \quad > \quad 0 = R(\{3,2\}) - R(\{2\})$
  - Submodularity requires: $R(A \cup \{3\}) - R(A) \leq R(B \cup \{3\}) - R(B)$

Absence of Structure

$$\max_{S \subseteq [n]} R(S)$$

where $R(S) = \sum_{i \in S} r_i \, \phi(i, S)$

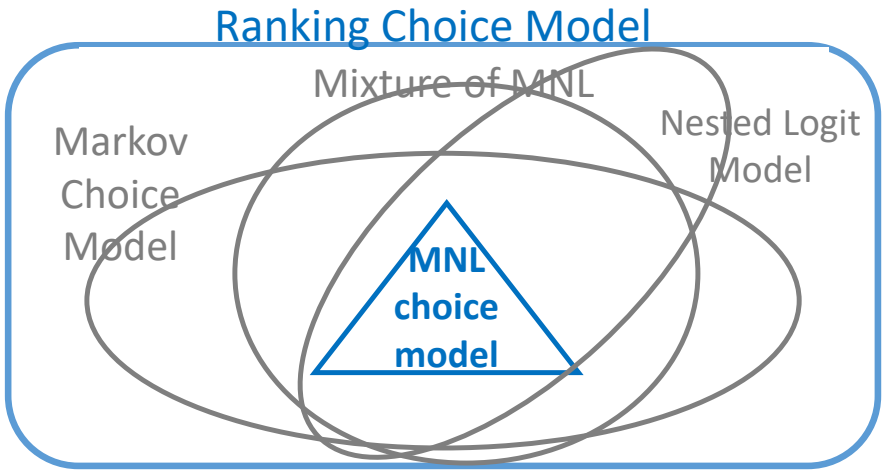
- Function is **not monotone, nor submodular**
- Problem is **very hard to even approximate**
  - Even without any constraints!

Do we need RCM?

- RCM is very expressive
- But also very hard to **learn** and **optimize**
- Practitioners often prefer simple models
  - We will also see real life application later

Multinomial Logit Model (MNL)

A very simple choice model



Multinomial Logit Choice (MNL)

v_j: weight of product $j \in [n]$

v_0: weight of outside option

$$\phi(i, S) = \frac{v_i}{v_0 + \sum_{j \in S} v_j}$$

- Examples: $\phi(\{1\}) = \frac{v_1}{v_0 + v_1}$, $\phi(\{1\}, \{1,2\}) = \frac{v_1}{v_0 + v_1 + v_2}$

Multinomial Logit Choice (MNL)

$$\phi(i,S) = \frac{v_i}{v_0 + \sum_{j \in S} v_j}$$

- **Optimization and learning are tractable (to be shown)**
- **Often the go-to model, but has clear limitations**

Drawback of MNL: IIA property

Independence of Irrelevant Alternatives: Given two products $i, j \in S$, probability that i is chosen over j is independent of S

$$\frac{\phi(i,S)}{\phi(j,S)} = \frac{v_i}{v_j}$$

IIA: Example

- The odds of i vs j are v_i/v_j, whatever else is on the menu

- Burger joint: cheeseburger $v_1=2$, veggie patty $v_2=1$, $v_0=1$
  - Offer {cheese, veggie}: odds are 0.5 vs 0.25 → chesse twice as likely
  - Add mushroom $v_3=1$: odds are 0.4 vs 0.2 → cheese still twice as likely

- Every product loses customers **in proportion** to its weight

Problem with IIA

Consider a small town with 3 burger joints

- McDonalds
  - 50 % of population prefers
- Burger King
  - 25 % of population prefers
- Shake Shack
  - 25 % of population prefers

- **What is the MNL model that captures this distribution?**

$$\left(\phi(i,S) = \frac{v_i}{v_0 + \sum_{j \in S} v_j}\right)$$

Problem with IIA

Consider a small town with 3 burger joints

- McDonalds
  - 50 % of population prefers
- Burger King
  - 25 % of population prefers
- Shake Shack
  - 25 % of population prefers

$$\phi(McD, \{McD, BK, SS\}) = \frac{50}{25 + 25 + 50} = 0.5$$

$$\phi(BK, \{McD, BK, SS\}) = \frac{25}{25 + 25 + 50} = 0.25$$

- **Representative MNL model:** $v_{McD} = 50 \;\; v_{BK} = 25, v_{SS} = 25, v_0 = 0$

Problem with IIA: When option disappears

Consider a small town with 3 burger joints

- McDonalds
  - 50 % of population prefers
- Burger King
  - 25 % of population prefers
- Shake Shack – closes
  - Customers substitute to other options
  - How?

$$\phi(McD, \{McD, BK\}) = \frac{50}{25+50} = \frac{2}{3}$$

$$\phi(BK, \{McD, BK\}) = \frac{25}{25+50} = \frac{1}{3}$$

IIA: $\frac{\phi(McD,S)}{\phi(BK,S)} = 2$

- Representative MNL model: $v_{McD} = 50 \;\; v_{BK} = 25, v_{SS} = 25, v_0 = 0$

Reality: Vegetarians went to SShack

- McDonalds
  - Still 50 % of population prefers — MNL says 67 %
    - No veggie options
- Burger King
  - 37.5 % of population prefers — MNL says 33 %
  - Half the vegetarians now go here
- Remaining 12.5 % (vegetarians) choose outside option — MNL says 0 %
- Shake Shack – closes
  - Customers substitute to other options

Does this make MNL useless?

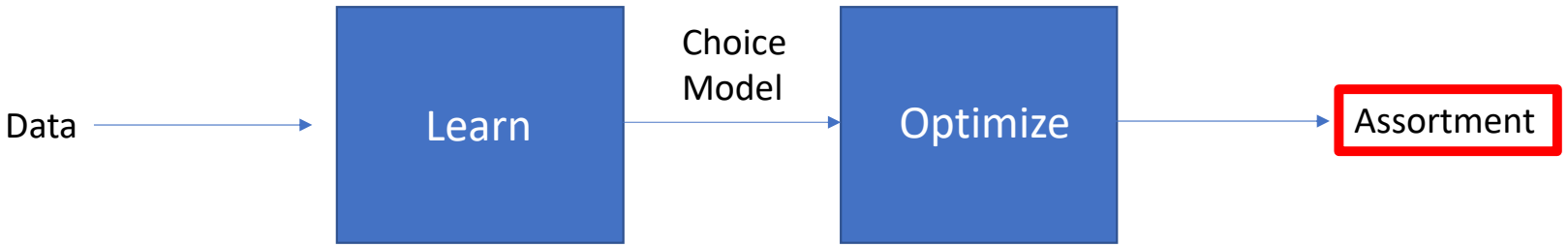
- MNL is useful in many applications
  - It captures the “first order” effect of substitution

- To quote a practitioner:

  “A good model can be explained to a 5th grader”

- Not always true but a good place to start

Data to Decision Pipeline



All steps are tractable for MNL choice

Revenue Optimization for MNL Choice

MNL Objective

$$R(S) = \sum_{i \in S} \frac{r_i v_i}{v_0 + \sum_{j \in S} v_j}$$

- Is $R(S)$ monotone?
  - No, there are examples with just 2 resources (Problem set 2)
- Is it submodular?
  - No, example with just 3 resources (Problem set 2)

Is it hard to Optimize with MNL too?

- It is easy but we **must use special structure of MNL**

- General property, such as submodularity, does not apply
  - There are extensions of submodularity that apply
  - But Greedy algorithm still performs poorly

- Solving linear/integer program is a common approach
  - Tractable formulations exist for various choice models including MNL (special cases of RCM/RUM)

Unconstrained MNL Optimization

MNL Optimization via Math. Programming

- $x_i \in \{0,1\}$ binary decision variable for product i

Non-linear: ratio of two linear terms

$$\max_{(x_i)_{i\in[n]}} \sum_{i\in[n]} r_i x_i \frac{v_i}{\underbrace{v_0 + \sum_{j\in[n]} v_j x_j}_{\phi(i,S_x)}}$$

Linearizing by Substitution and Relaxation

$$\max_{x_i \in \{0,1\}\ \forall\, i \in [n]} \sum_{i \in [n]} r_i x_i \frac{v_i}{v_0 + \sum_{j \in [n]} v_j x_j}$$

$x_i = \frac{y_i}{\theta}$

$$\max_{(y_i)_{i \in [n]}, \theta} \sum_{i \in [n]} r_i v_i \ y_i$$

$$s.t. \qquad 0 \le y_i \le \theta \ \ \forall\, i$$

$$v_0 \theta + \sum_{j \in [n]} v_j y_j = 1$$

$$\theta \ge 0$$

Up next

Linearizing by Substitution

- $x_i \in \{0,1\}$ binary decision variable for product i

$$\left(\sum_{i\in[n]} r_i x_i \frac{v_i}{v_0 + \sum_{j\in[n]} v_j x_j}\right)$$

$$\max_{\theta,\, x_i} \sum_{i\in[n]} r_i v_i \, x_i \, \theta$$

$$s.t. \;\; \theta = \frac{1}{v_0 + \sum_{j\in[n]} v_j x_j}$$

$$\theta \geq 0$$

Linearizing by Substitution

- $x_i \in \{0,1\}$ binary decision variable for product i

Still non-linear

$$\max_{\theta,\, x_i} \sum_{i \in [n]} r_i v_i \; x_i \, \theta$$

$$s.t. \;\; \theta = \frac{1}{v_0 + \sum_{j \in [n]} v_j x_j}$$

$$\theta \geq 0$$

Another Substitution

$$\left(\max_{\theta,\, x_i} \sum_{i\in[n]} r_i v_i \;\; x_i\, \theta\right)$$

$$\max_{\theta, x_i, y_i} \sum_{i\in[n]} r_i v_i \;\; y_i$$

$$s.t. \qquad y_i = x_i\, \theta \;\; \forall\, i$$

$$\theta = \frac{1}{v_0 + \sum_{j\in[n]} v_j x_j}$$

$$x_i \in \{0,1\}, \theta \geq 0$$

Another Substitution

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$$v_0 \theta + \sum_{j \in [n]} v_j y_j = 1$$

$$x_i \in \{0,1\}, \theta \geq 0$$

Another Substitution: An Integer Program

$$\max_{\theta, \sout{x_i}, y_i} \sum_{i \in [n]} r_i v_i \; y_i$$

$$s.t. \qquad y_i \in \{0, \theta\} \;\; \forall \; i$$

Because x_i does not appear elsewhere

$$\left(\theta = \frac{1}{v_0 + \sum_{j \in [n]} v_j x_j} \right)$$

$$v_0 \theta + \sum_{j \in [n]} v_j y_j = 1$$

$$\sout{x_i \in \{0,1\}}, \theta \geq 0$$

LP Relaxation

$$\max_{\theta, y_i} \sum_{i \in [n]} r_i v_i \; y_i$$

$(y_i = x_i \theta)$ $\quad s.t. \quad 0 \leq y_i \leq \theta \;\; \forall \, i$

$$v_0 \theta + \textstyle\sum_{j \in [n]} v_j y_j = 1$$

$$x_i \in [0,1], \theta \geq 0$$

Optimizing MNL Revenue (Unconstrained)

- Solve:

$$\max_{(y_i)_{i\in[n]},\theta} \sum_{i\in[n]} r_i v_i \; y_i$$

$$s.t. \qquad 0 \le y_i \le \theta \;\; \forall\, i$$

$$v_0\theta + \textstyle\sum_{j\in[n]} v_j y_j = 1$$

$$\theta \ge 0$$

- Read the assortment back with $x_i = y_i/\theta$
  - x_i must be 0,1 so will y_i/θ always be integer?
  - Yes! Proof beyond our scope (Total unimodularity shows that LP always has integer optimal)

Cardinality Constrained
MNL optimization

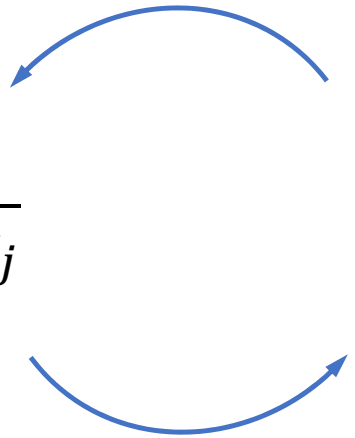
MNL with cardinality constraint

$$\max_{|S_x| \leq k} \sum_{i \in S} \frac{r_i v_i x_i}{v_0 + \sum_{j \in S} v_j x_j}$$

LP Approach

$$\max_{\sum_i x_i \le k} \sum_{i \in S} \frac{r_i v_i x_i}{v_0 + \sum_{j \in S} v_j x_j}$$

$x_i = \frac{y_i}{\theta}$



$$\max_{y_i, \theta} \sum_{i \in [n]} r_i v_i \ y_i$$

$$s.t. \ \sum_i \frac{y_i}{\theta} \le k$$

$$v_0 \theta + \sum_{j \in [n]} v_j y_j = 1$$

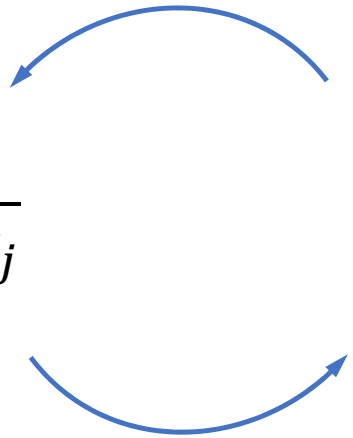
$$0 \le y_i \le \theta$$

$$\theta \ge 0$$

LP Approach

$$\max_{\sum_i x_i \le k} \sum_{i \in S} \frac{r_i v_i x_i}{v_0 + \sum_{j \in S} v_j x_j}$$

$x_i = \frac{y_i}{\theta}$



$$\max_{y_i, \theta} \sum_{i \in [n]} r_i v_i \; y_i$$

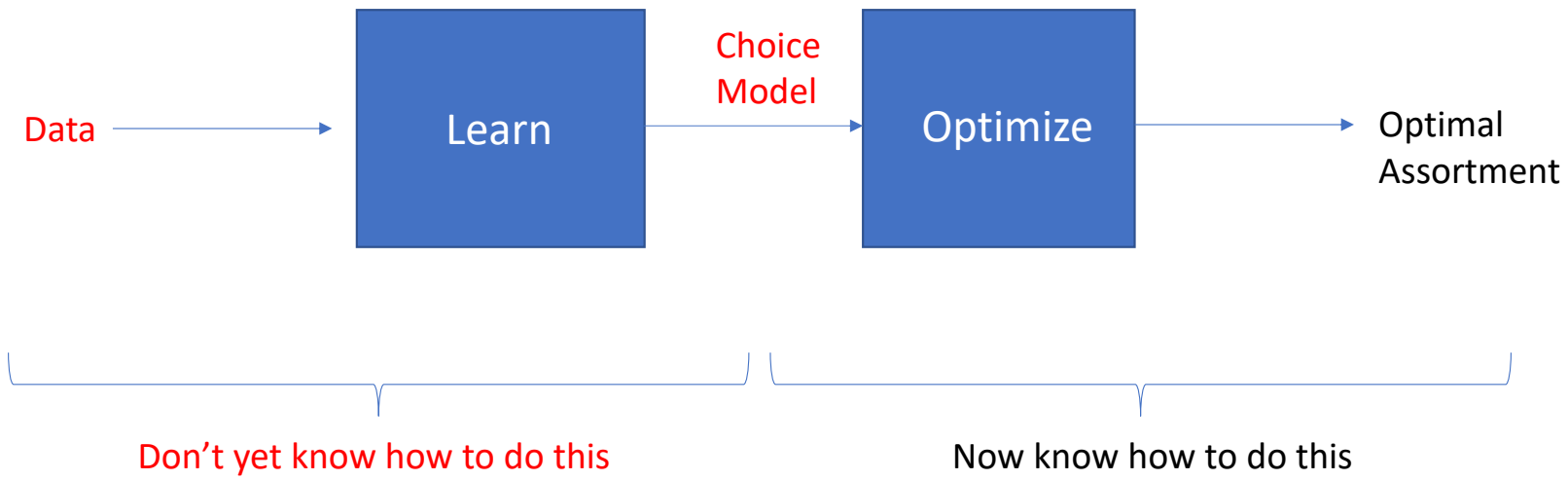
$$s.t. \quad \sum_i y_i - k\,\theta \le 0$$

$$v_0 \theta + \sum_{j \in [n]} v_j y_j = 1$$

$$0 \le y_i \le \theta$$

$$\theta \ge 0$$

From Data to Decision



Learning MNL Choice Model

Learning a Distribution

- Suppose you have a biased coin
  - You don't know the bias
- You flip it 10 times and see the following:

  H H H H H T H T H H
- What is your estimate of the probability of heads?

Learning a Distribution

- Suppose you have a biased coin
  - You don't know the bias
- You flip it 10 times and see the following:

  H H H H H T H T H H
- What is your estimate of the probability of heads?
  - $\frac{8}{10}$ – is there a systematic way to get this?

Maximum Likelihood Estimate

- Suppose probability of heads is p
- Then, what is probability of getting H H H H H T H T H H ?

$$\text{Likelihood}(\text{p}) = p^8(1-p)^2$$

Maximum Likelihood Estimate

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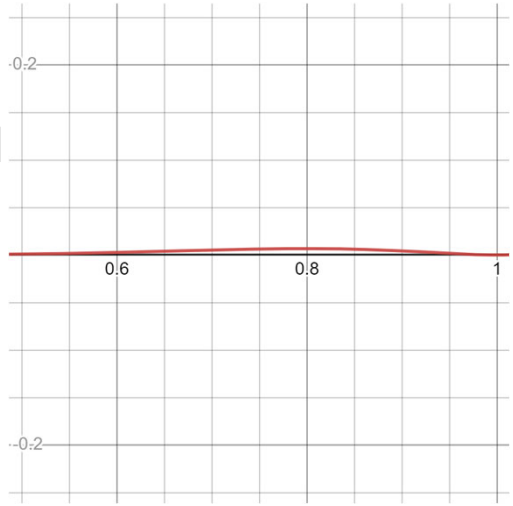
- What value of p maximizes likelihood?

$$p^* = \text{argmax}_{p\in[0,1]}\, p^8(1-p)^2$$

Maximum Likelihood Estimate

- Suppose probability of heads is p
- Then, what is probability of getting H H H H H T H T H H

$$\text{Likelihood(p)} = p^8(1-p)^2$$

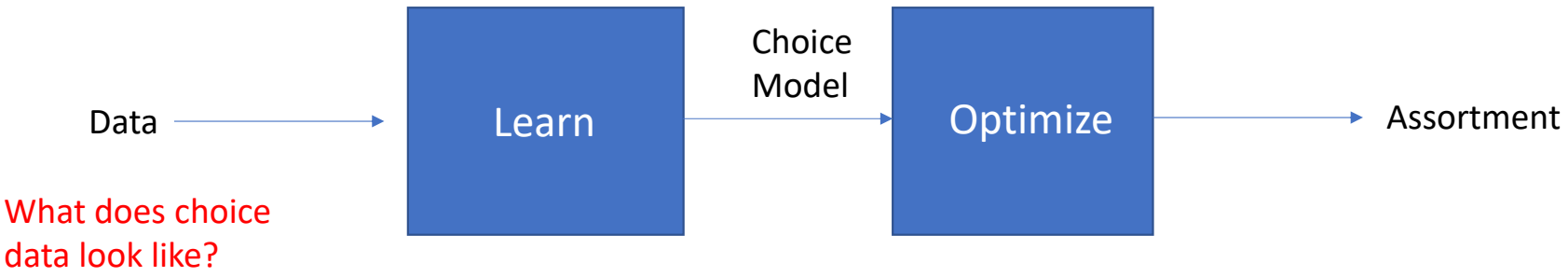


- What value of p maximizes likelihood?

$$p^* = \text{argmax}_{p\in[0,1]}\, p^8(1-p)^2 = \frac{8}{10}$$

Maximum Likelihood Estimate (MLE)

From Data to Decision



What Would Data Look Like?

- Typical sample - assortment shown + choice made

S_t, i_t: customer t saw assortment S_t and chose $i_t \in S_t \cup \{0\}$

- Available in brick and mortar stores as well as online platforms
  - Online platforms also have other data such as “clicks” or “views”

What Would Data Look Like?

- Typical sample consists of the assortment shown and choice made

$$S_t, i_t\text{: customer } t \text{ saw assortment } S_t \text{ and chose } i_t \in S_t \cup \{0\}$$

- Available in brick and mortar stores as well as online platforms
  - Online platforms also have other data such as “clicks” or “hover”

- Dataset is a collection of **independent** samples $\{S_t, i_t\}_{t \in T}$
  - Called *sales data*

Sales Data: Examples

- A café's log, one row per customer t:
  - Customer 1 saw {latte, tea}, chose latte → $(S_1, i_1) = (\{1,2\}, 1)$
  - Customer 2 saw {latte, tea, cold brew}, left → $(\{1,2,3\}, 0)$
  - Customer 3 saw {tea, cold brew}, chose tea → $(\{2,3\}, 2)$
- Online: the products on the page and the one put in the cart
- **Independent**: each customer chooses on their own; one sample does not affect another

Given a Single Sample S_t, i_t

Example: $\{1,2,3\}, 1$

- On ground set $N = \{1,2,3\}$, are all MNL models viable for this sample?

Given a Single Sample S_t, i_t

Example: $\{1,2,3\}, 1$

- Are all MNL models viable for this sample?
- Models with $v_1 = 0$ are not viable
- Most likely model is,

$$\text{argmax}_{(v_1,v_2,v_3)}\ \phi(1,\{1,2,3\}) = \text{argmax}_{(v_1,v_2,v_3)}\ \frac{v_1}{1+v_1+v_2+v_3}$$

($v_0 = 1$ as part of normalization)